Payout Earnings Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Payout Earnings Sensitivity (PES), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on PES achieves an annualized gross (net) Sharpe ratio of 0.30 (0.25), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 29 (20) bps/month with a t-statistic of 3.19 (2.28), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities) is 20 bps/month with a t-statistic of 2.18.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, yet identifying which firm characteristics capture this risk exposure remains a central challenge in finance. While the consumption-based asset pricing model provides a theoretical foundation for understanding expected returns through the lens of marginal utility, empirical work continues to uncover cross-sectional patterns that require deeper economic interpretation. We identify a novel predictor of stock returns based on the sensitivity of firm payouts to earnings changes, which we term Payout Earnings Sensitivity (PES). This measure captures firms' differential exposure to aggregate consumption risk through their payout policies during varying economic states. Firms with high payout-earnings sensitivity demonstrate greater resilience in maintaining shareholder distributions during earnings shocks, suggesting lower exposure to systematic consumption risk and thus commanding lower expected returns in equilibrium.

The theoretical foundation for PES emerges from the consumption-based asset pricing framework, where expected returns compensate investors for covariance with marginal utility growth Lucas (1978); Breeden (1979). Firms exhibiting high payout-earnings sensitivity maintain stable distributions to shareholders even when earnings fluctuate, effectively smoothing consumption streams for their investors. This smoothing capability becomes particularly valuable during economic downturns when marginal utility is high, as documented in models of long-run risks Bansal and Yaron (2004) and rare disasters Barro (2006). The ability to sustain payouts despite earnings volatility suggests these firms face lower operating leverage and possess greater financial flexibility, characteristics that reduce their exposure to aggregate consumption risk.

The state-dependent nature of payout policy directly connects to the intertemporal marginal rate of substitution that prices all assets in equilibrium. During bad economic states characterized by high marginal utility, firms that can maintain dividend payments provide valuable consumption hedging to investors Campbell and Cochrane (1999). These firms' cash flows exhibit lower correlation with the stochastic discount factor, particularly in states where consumption growth is low or negative. The habit formation models of Constantinides (1990) further suggest that investors place premium value on assets that help smooth consumption relative to habit levels, implying lower required returns for high-PES firms that facilitate this smoothing.

The economic mechanism linking PES to consumption risk operates through firms' differential exposure to aggregate productivity shocks. High-PES firms demonstrate operational characteristics that insulate their distributable cash flows from earnings volatility, potentially through diversified revenue streams, lower fixed costs, or superior working capital management Zhang (2005). This insulation manifests in their ability to maintain payout ratios when earnings decline, providing consumption insurance precisely when investors' marginal utility peaks. The cross-sectional variation in PES thus captures heterogeneous loadings on consumption-based risk factors, with low-PES firms requiring higher expected returns to compensate investors for their procyclical payout patterns that amplify consumption risk during economic contractions.

We find strong evidence that PES predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on PES achieving an annualized gross Sharpe ratio of 0.30 and a net Sharpe ratio of 0.25 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 29 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 3.19. This performance remains economically and statistically significant after controlling for the six most closely related strategies from the factor zoo, yielding a gross monthly alpha of 20 basis points with a t-statistic of 2.18.

The economic magnitude of the PES effect manifests consistently across different portfolio construction methodologies and size groups. Among the largest stocks, those with market capitalization above the 80th NYSE percentile, the PES strategy achieves an average return of 14 basis points per month, demonstrating that the effect is not confined to small, illiquid securities. The strategy's factor loadings reveal a significant negative exposure to the CMA factor of -0.79 with a t-statistic of -12.57, consistent with high-PES firms maintaining conservative asset growth that preserves payout flexibility.

Our results demonstrate remarkable robustness to transaction costs and implementation considerations. The net Sharpe ratio of 0.25 places the PES strategy in the top 17% of anomalies in the factor zoo, while the strategy's net generalized alphas remain positive across all five standard factor models tested. After accounting for trading costs, a dollar invested in the PES strategy would have grown to $2.19, ranking in the top 7% of all documented anomalies. These findings establish PES as an economically important predictor that expands the investable opportunity set even for sophisticated investors with access to established factor models.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While Fama and French (2015) document that profitability and investment patterns predict returns, and Hou et al. (2015) develop a related investment-based factor model, our PES measure provides a direct link between firm payout policies and consumption risk that these models do not fully capture. Unlike the mechanical accounting relationships underlying many profitability measures, PES reflects managers' forward-looking assessments of their firms' ability to sustain distributions through varying economic conditions, providing unique information about consumption risk exposure.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023), which addresses concerns about data mining and implementation costs that plague many anomaly studies. Our analysis demonstrates that PES not only survives stringent robustness tests but also maintains economic significance after accounting for transaction costs, a hurdle that eliminates many documented anomalies as shown by Chen and Velikov (2022). The persistence of the PES effect across size groups and portfolio construction methods distinguishes it from anomalies that derive their apparent strength from small, illiquid stocks that are difficult to trade profitably.

The broader implications of our findings extend to corporate finance and macroeconomics, where understanding the determinants of payout policy remains an active area of research. By demonstrating that payout-earnings sensitivity predicts returns through a consumption risk channel, we provide new evidence on the real economic forces that shape optimal payout policy. Our results suggest that firms' payout flexibility serves as a valuable characteristic for investors seeking to hedge consumption risk, with equilibrium prices reflecting this hedging value through lower expected returns. This insight contributes to the growing literature linking corporate policies to asset prices through their implications for investors' consumption and welfare.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on firms with sufficient data to calculate year-over-year changes in operating performance and dividend payment history.construction of our Payout Earnings Sensitivity signal requires two key variables from COMPUSTAT. Operating income after depreciation (OIADP) represents a firm's operating income after accounting for depreciation and amortization expenses, providing a measure of operating performance that reflects the cost of using long-term assets. Total dividends (DVT) captures the total amount of cash dividends declared on all equity capital, including dividends on common stock, preferred stock, and any other equity instruments. Both variables are measured in millions of dollars and reported on an annual basis.construct the Payout Earnings Sensitivity signal by calculating the year-over-year change in operating income after depreciation, scaled by lagged total dividends. Specifically, we compute (OIADP(t) - OIADP(t-1)) / DVT(t-1), where t denotes the current fiscal year. This ratio captures the sensitivity of a firm's operating performance changes relative to its dividend commitment level, providing insight into how earnings fluctuations relate to the firm's existing payout obligations. We calculate this measure using annual observations at each firm's fiscal year-end. To ensure data quality, we require non-missing values for operating income after depreciation in consecutive years and positive lagged dividends to avoid division by zero. Firms with zero or missing dividend values in the denominator are excluded from the analysis for that year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PES signal. Panel A plots the time-series of the mean, median, and interquartile range for PES. On average, the cross-sectional mean (median) PES is 4.00 (0.44) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PES data. The signal's interquartile range spans -2.92 to 3.70. Panel B of Figure 1 plots the time-series of the coverage of the PES signal for the CRSP universe. On average, the PES signal is available for 3.66% of CRSP names, which on average make up 6.86% of total market capitalization.

4 Does PES predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PES using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PES portfolio and sells the low PES portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PES strategy earns an average return of 0.27% per month with a t-statistic of 2.35. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 1.60 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.79, with a t-statistic of -12.57 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 323 stocks and an average market capitalization of at least $858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -3-28bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PES trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in ten cases.

Table 3 provides direct tests for the role size plays in the PES strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PES, as well as average returns and alphas for long/short trading PES strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the PES strategy achieves an average return of 14 bps/month with a t-statistic of 1.11. Among these large cap stocks, the alphas for the PES strategy relative to the five most common factor models range from 3 to 24 bps/month with t-statistics between 0.21 and 2.25.

5 How does PES perform relative to the zoo?

Figure 2 puts the performance of PES in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the PES strategy falls in the distribution. The PES strategy's gross (net) Sharpe ratio of 0.30 (0.25) is greater than 65% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the PES strategy (red line).[2] Ignoring trading costs, a $1 invested in the PES strategy would have yielded $3.47 which ranks the PES strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a $1 invested in the PES strategy would have yielded $2.19 which ranks the PES strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PES relative to those. Panel A shows that the PES strategy gross alphas fall between the 36 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PES strategy has a positive net generalized alpha for five out of the five factor models. In these cases PES ranks between the 54 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PES add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PES with 205 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PES or at least to weaken the power PES has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES} PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PES and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PES signal in these Fama-MacBeth regressions exceed 0.59, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on PES is 2.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the PES strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PES strategy earns alphas that range from 21-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.26, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PES trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.18.

7 Does PES add relative to the whole zoo?

Finally, we can ask how much adding PES to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PES signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes PES grows to $2576.57.

8 Conclusion

This study provides compelling evidence for the predictive power of Payout Earnings Sensitivity (PES) in the cross-section of equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PES is available.

that PES represents a economically meaningful and statistically robust signal for predicting stock returns. The value-weighted long/short strategy based on PES delivers substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.25 and significant alpha relative to both the Fama-French five-factor model plus momentum (20 bps/month, t-statistic = 2.28) and an extended model including six closely related factors (20 bps/month, t-statistic = 2.18). These results suggest that PES captures a distinct source of cross-sectional return variation not explained by existing factor models or related growth and investment-based strategies. The economic magnitude of the signal, combined with its statistical robustness after accounting for transaction costs, indicates practical relevance for portfolio construction and risk management. The persistence of significant alpha even after controlling for conceptually similar factors such as asset growth, employment growth, and changes in operating liabilities suggests that PES contains unique information about firms' fundamental characteristics and their implications for expected returns. However, several limitations warrant consideration. First, while our net returns account for transaction costs, actual implementation may face additional frictions including market impact and capacity constraints. Second, the sample period and market conditions examined may not fully capture the signal's performance across different economic regimes. Future research should investigate the economic mechanisms underlying the PES effect, potentially exploring whether it reflects systematic risk exposure, behavioral biases, or market frictions. Additionally, examining the signal's performance in international markets would provide valuable out-of-sample validation. Researchers might also investigate potential enhancements through machine learning techniques or by combining PES with other predictive signals. Finally, understanding the time-varying nature of the PES premium and its relationship to macroeconomic conditions could yield important insights for dynamic portfolio strategies.

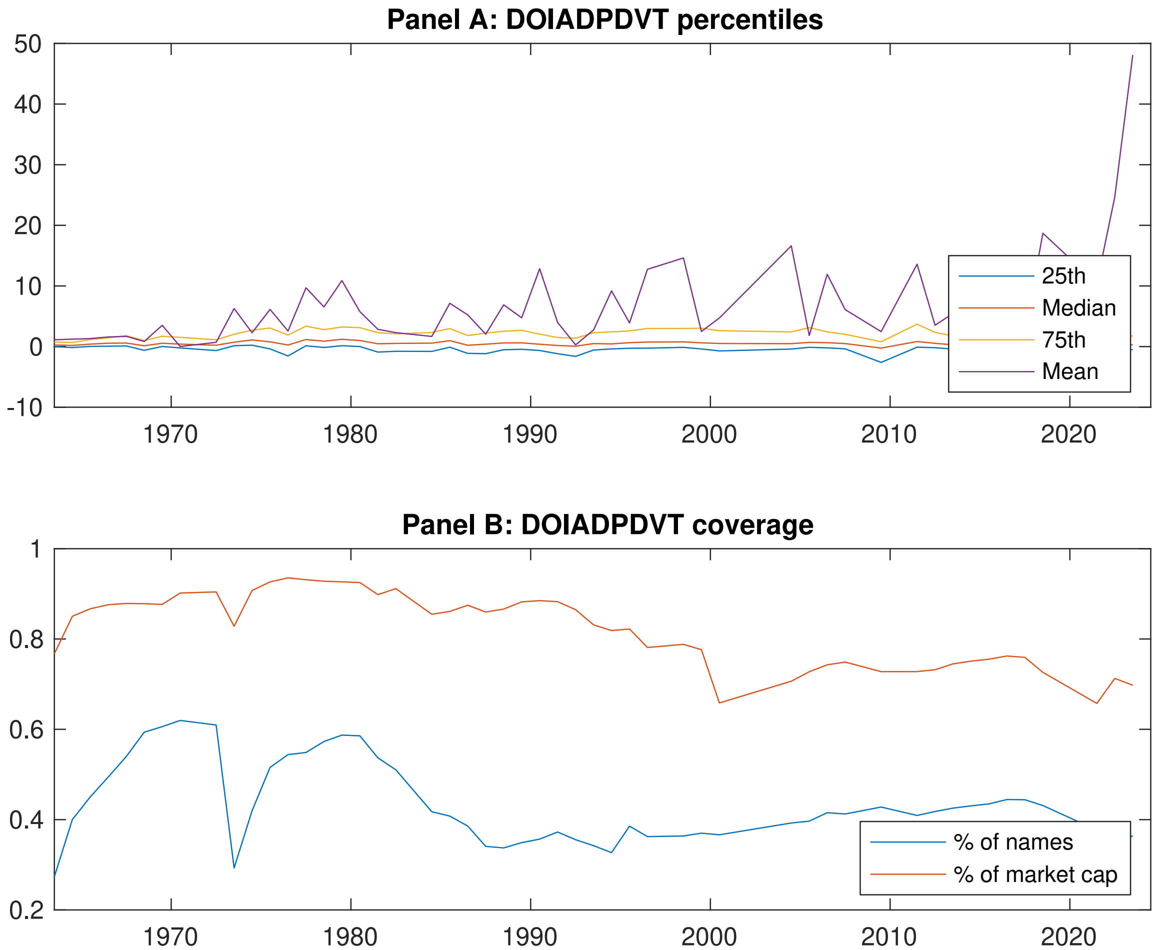


Figure 1: Times series of PES percentiles and coverage.
This figure plots descriptive statistics for PES. Panel A shows cross-sectional percentiles of PES over the sample. Panel B plots the monthly coverage of PES relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PES. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PES-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44	0.58	0.60	0.68	0.72	0.27
	[2.32]	[3.83]	[4.08]	[4.05]	[3.40]	[2.35]
α_{CAPM}	-0.17	0.10	0.12	0.11	0.01	0.18
	[-2.18]	[1.43]	[2.06]	[2.14]	[0.11]	[1.60]
α_{FF3}	-0.31	0.01	0.08	0.09	0.02	0.33
	[-4.47]	[0.14]	[1.62]	[1.79]	[0.32]	[3.16]
α_{FF4}	-0.26	-0.00	0.07	0.09	0.05	0.31
	[-3.66]	[-0.07]	[1.49]	[1.78]	[0.67]	[2.83]
α_{FF5}	-0.31	-0.14	-0.05	0.01	0.02	0.33
	[-4.63]	[-2.67]	[-0.99]	[0.25]	[0.30]	[3.62]
α_{FF6}	-0.26	-0.13	-0.04	0.02	0.03	0.29
	[-3.86]	[-2.52]	[-0.86]	[0.34]	[0.55]	[3.19]
Panel B: Fama and French (2018) 6-factor model loadings for PES-sorted portfolios						
β_{MKT}	1.09	0.95	0.91	1.00	1.14	0.05
	[67.71]	[76.42]	[81.90]	[90.50]	[75.03]	[2.37]
β_{SMB}	0.08	-0.15	-0.14	-0.04	0.14	0.06
	[3.47]	[-8.30]	[-8.79]	[-2.66]	[6.56]	[2.00]
β_{HML}	0.18	0.06	0.09	0.12	0.11	-0.07
	[5.75]	[2.39]	[4.33]	[5.82]	[3.76]	[-1.61]
β_{RMW}	-0.17	0.13	0.29	0.28	0.27	0.43
	[-5.36]	[5.42]	[13.38]	[12.96]	[8.93]	[10.07]
β_{CMA}	0.35	0.52	0.12	-0.10	-0.44	-0.79
	[7.63]	[14.69]	[3.87]	[-3.32]	[-10.15]	[-12.57]
β_{UMD}	-0.07	-0.01	-0.01	-0.01	-0.02	0.05
	[-4.60]	[-0.69]	[-0.67]	[-0.57]	[-1.49]	[2.32]
Panel C: Average number of firms (n) and market capitalization (me)						
n	424	329	323	358	434	
me ($\$10^6$)	858	1725	1962	2021	1131	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the PES strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.18	0.33	0.31	0.33	0.29
			[2.35]	[1.60]	[3.16]	[2.83]	[3.62]	[3.19]
Quintile	NYSE	EW	0.15	0.12	0.17	0.07	0.08	-0.01
			[2.10]	[1.61]	[2.43]	[0.95]	[1.19]	[-0.19]
Quintile	Name	VW	0.33	0.25	0.39	0.35	0.37	0.32
			[2.74]	[2.07]	[3.50]	[3.05]	[3.78]	[3.26]
Quintile	Cap	VW	0.15	0.04	0.19	0.20	0.28	0.27
			[1.35]	[0.33]	[1.90]	[1.92]	[3.24]	[3.05]
Decile	NYSE	VW	0.31	0.23	0.38	0.32	0.33	0.28
			[2.16]	[1.62]	[2.77]	[2.33]	[2.74]	[2.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.12	0.26	0.24	0.22	0.20
			[1.96]	[1.05]	[2.45]	[2.30]	[2.47]	[2.28]
Quintile	NYSE	EW	-0.03					
			[-0.41]					
Quintile	Name	VW	0.28	0.18	0.32	0.29	0.26	0.24
			[2.33]	[1.52]	[2.81]	[2.59]	[2.71]	[2.47]
Quintile	Cap	VW	0.11		0.12	0.13	0.17	0.17
			[1.01]		[1.24]	[1.29]	[1.98]	[1.93]
Decile	NYSE	VW	0.25	0.15	0.29	0.26	0.22	0.19
			[1.74]	[1.07]	[2.11]	[1.88]	[1.82]	[1.58]

Table 3: Conditional sort on size and PES

This table presents results for conditional double sorts on size and PES. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PES. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PES and short stocks with low PES .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		PES Quintiles					PES Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.61	1.00	0.87	0.94	0.84	0.23	0.24	0.26	0.16	0.13	0.05
		[2.26]	[3.62]	[4.73]	[4.61]	[3.36]	[2.03]	[2.11]	[2.29]	[1.41]	[1.17]	[0.45]
	(2)	0.73	0.73	0.87	0.89	0.85	0.11	0.06	0.13	0.07	0.03	-0.02
		[3.18]	[3.98]	[4.93]	[4.34]	[3.49]	[1.17]	[0.59]	[1.37]	[0.69]	[0.36]	[-0.20]
	(3)	0.64	0.66	0.82	0.87	0.87	0.22	0.15	0.22	0.15	0.17	0.11
		[3.05]	[4.03]	[4.70]	[4.52]	[3.75]	[2.13]	[1.44]	[2.15]	[1.47]	[1.69]	[1.07]
	(4)	0.59	0.69	0.75	0.84	0.63	0.04	-0.03	0.04	-0.04	-0.07	-0.14
		[2.95]	[4.24]	[4.56]	[4.49]	[2.85]	[0.40]	[-0.30]	[0.38]	[-0.36]	[-0.71]	[-1.42]
	(5)	0.46	0.56	0.55	0.59	0.60	0.14	0.03	0.17	0.19	0.24	0.24
		[2.63]	[3.89]	[3.63]	[3.54]	[2.96]	[1.11]	[0.21]	[1.40]	[1.53]	[2.25]	[2.20]

Panel B: Portfolio average number of firms and market capitalization

		PES Quintiles					PES Quintiles				
		Average n					Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	149	152	154	153	152	11	13	14	14	14
	(2)	63	63	63	64	63	26	28	28	28	28
	(3)	54	54	54	54	54	55	56	58	58	56
	(4)	51	51	51	51	51	137	139	140	142	136
	(5)	53	54	53	54	53	1145	1425	1408	1404	1132

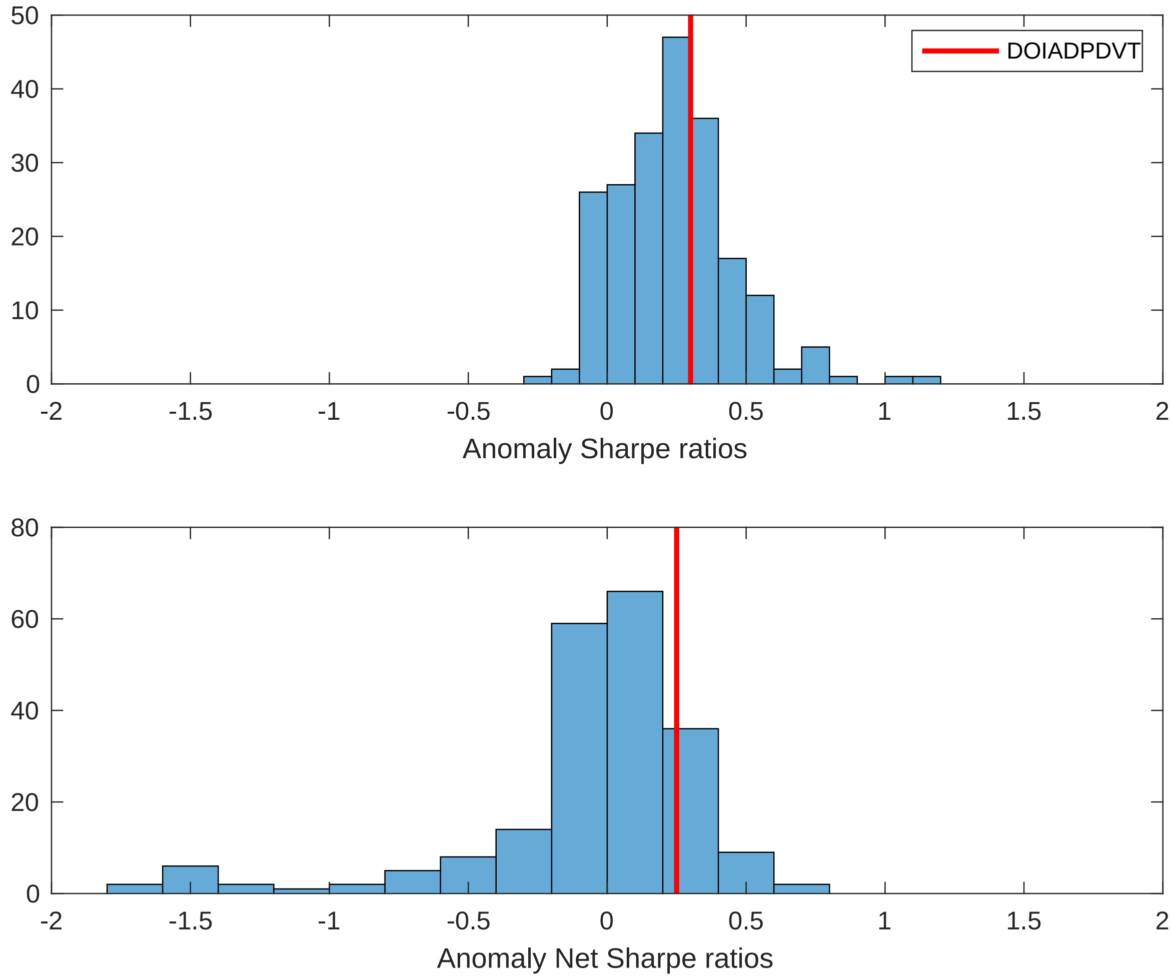


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PES with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

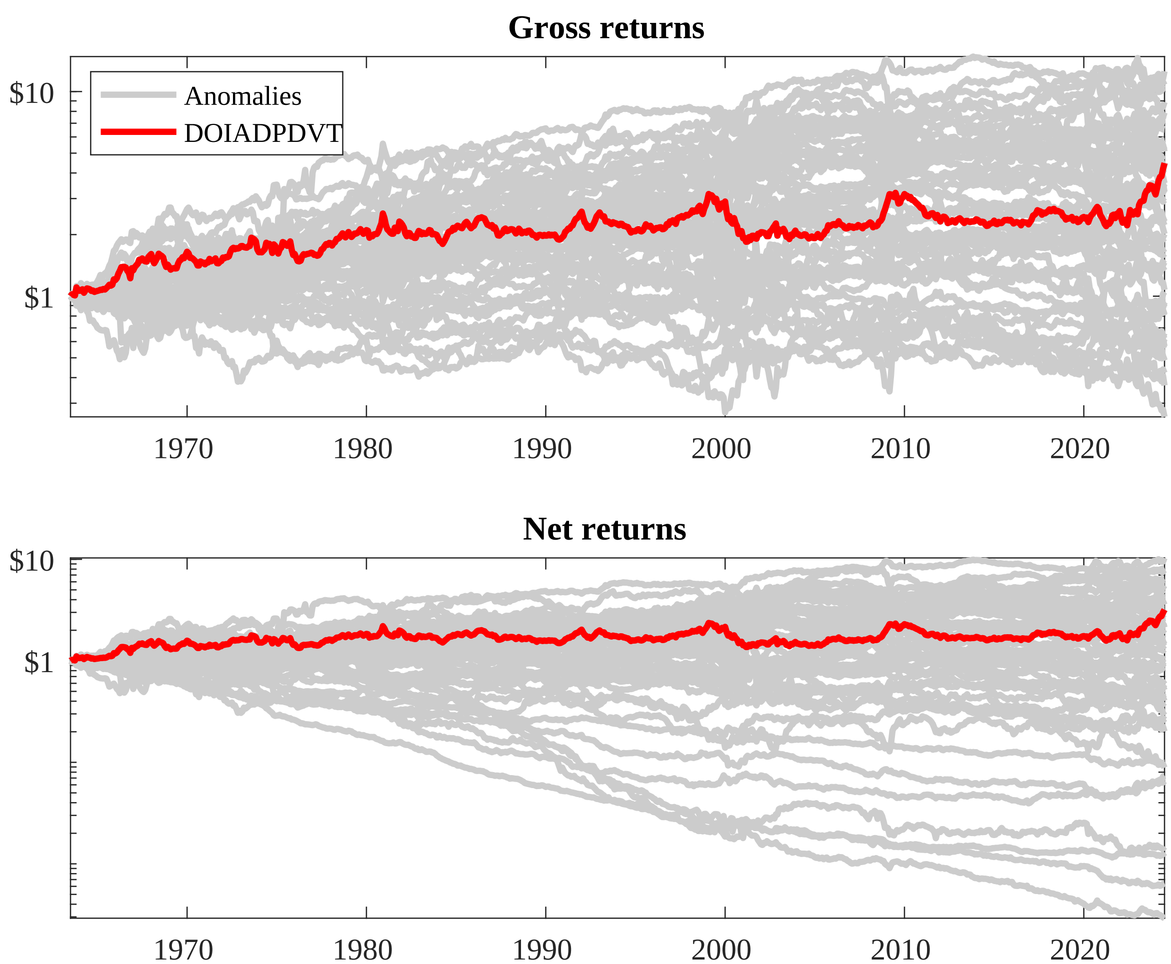
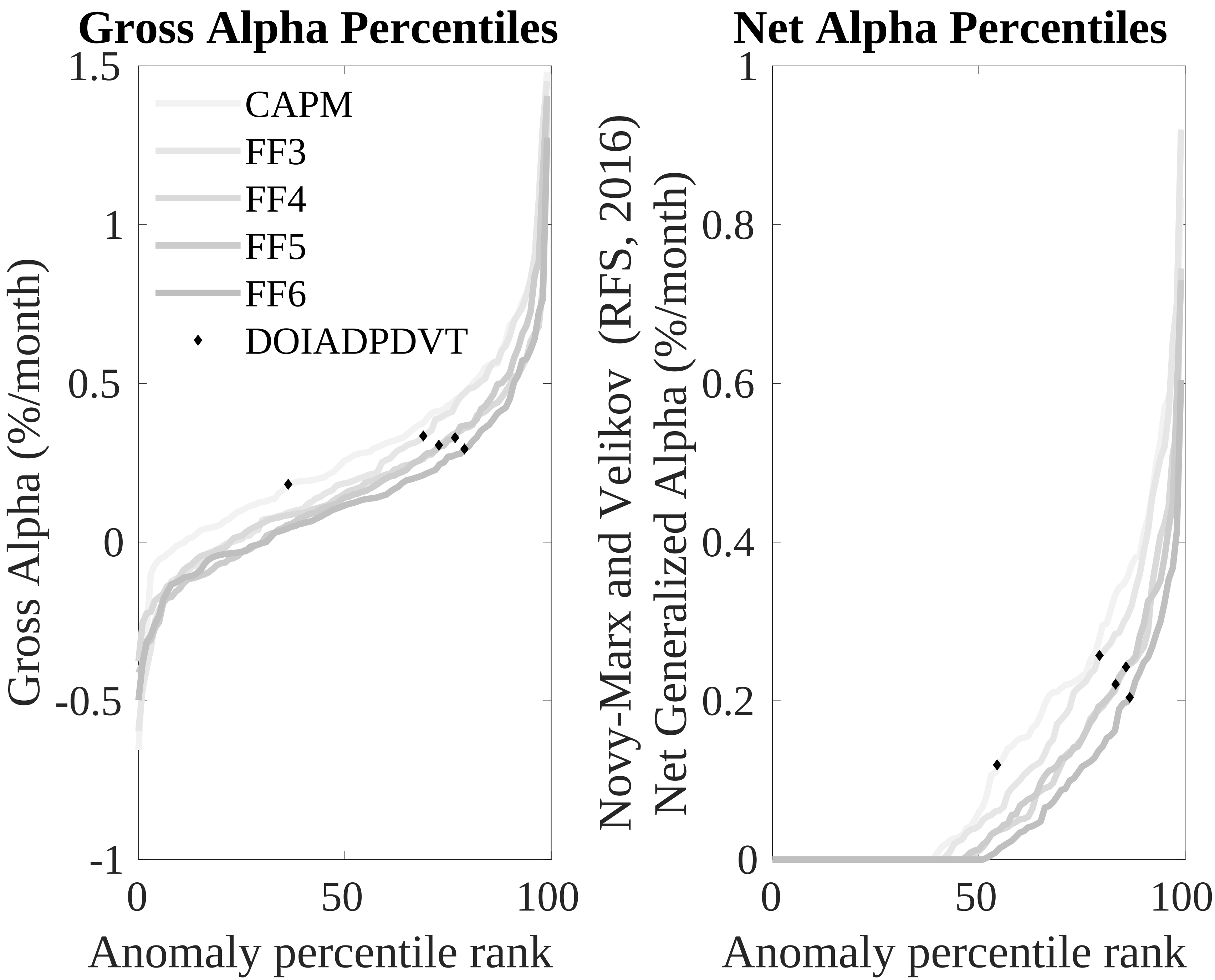
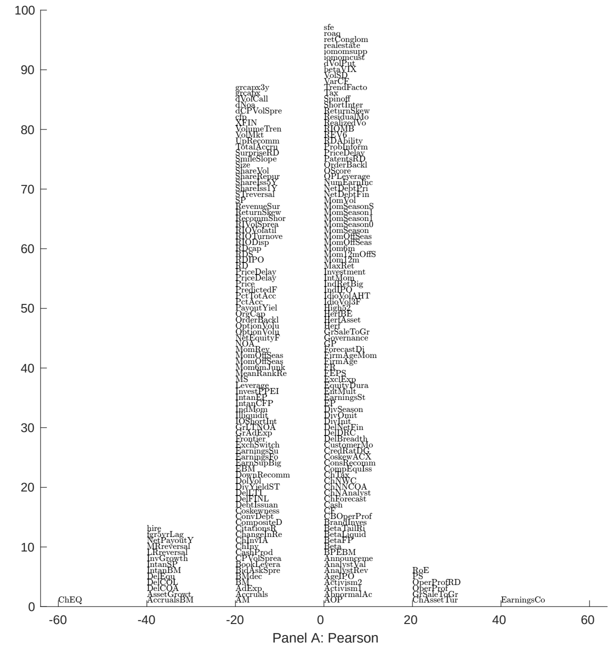


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PES trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
DOIADPDVT
1.5
1
0.5
0
-0.5
-1
0
50
100
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
1
0.8
0.6
0.4
0.2
0
0
50
100



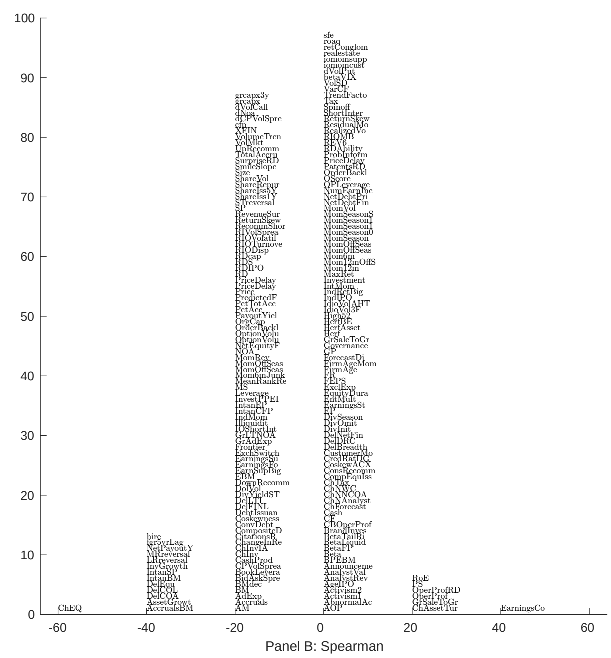


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 205 filtered anomaly signals with PES. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

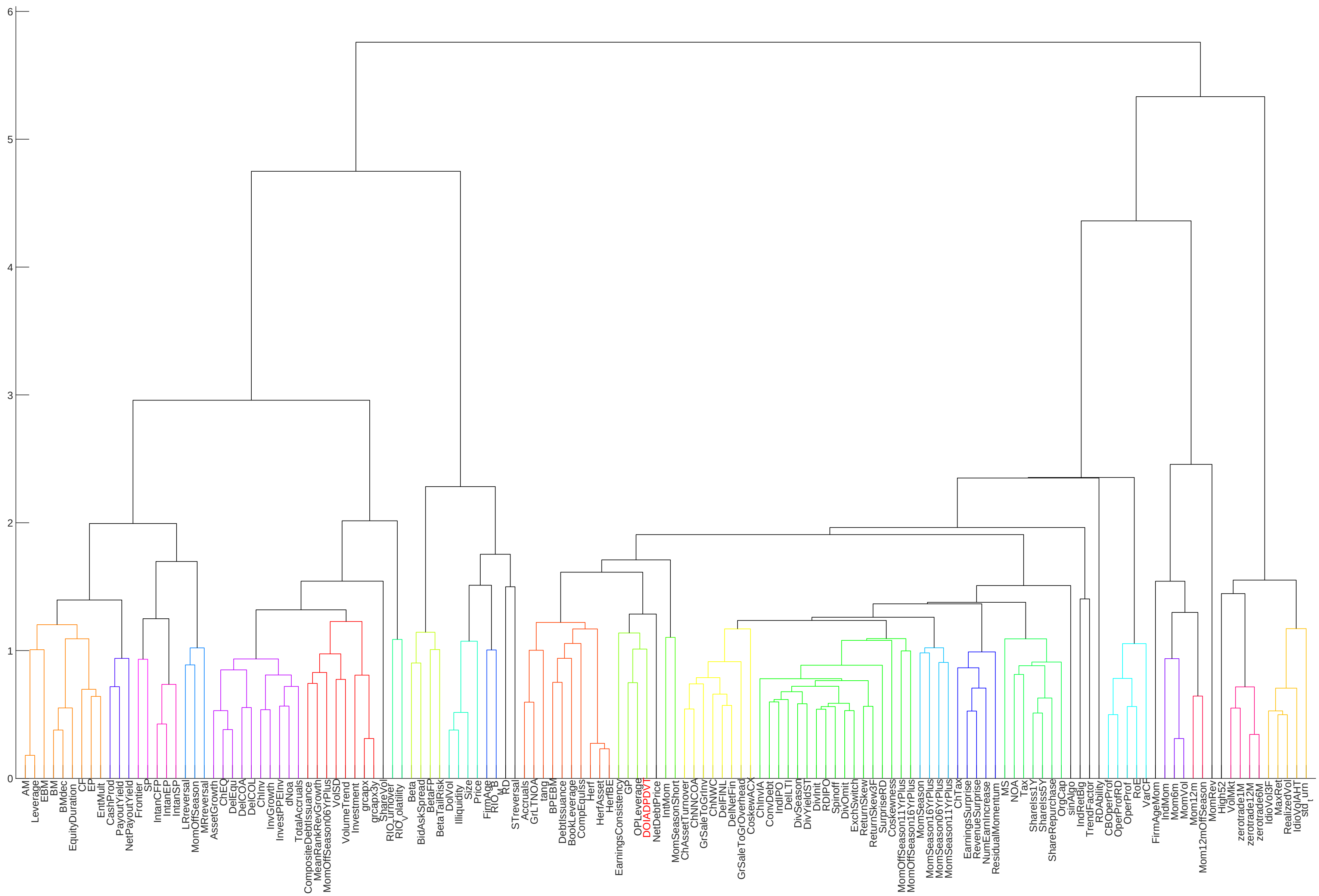


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

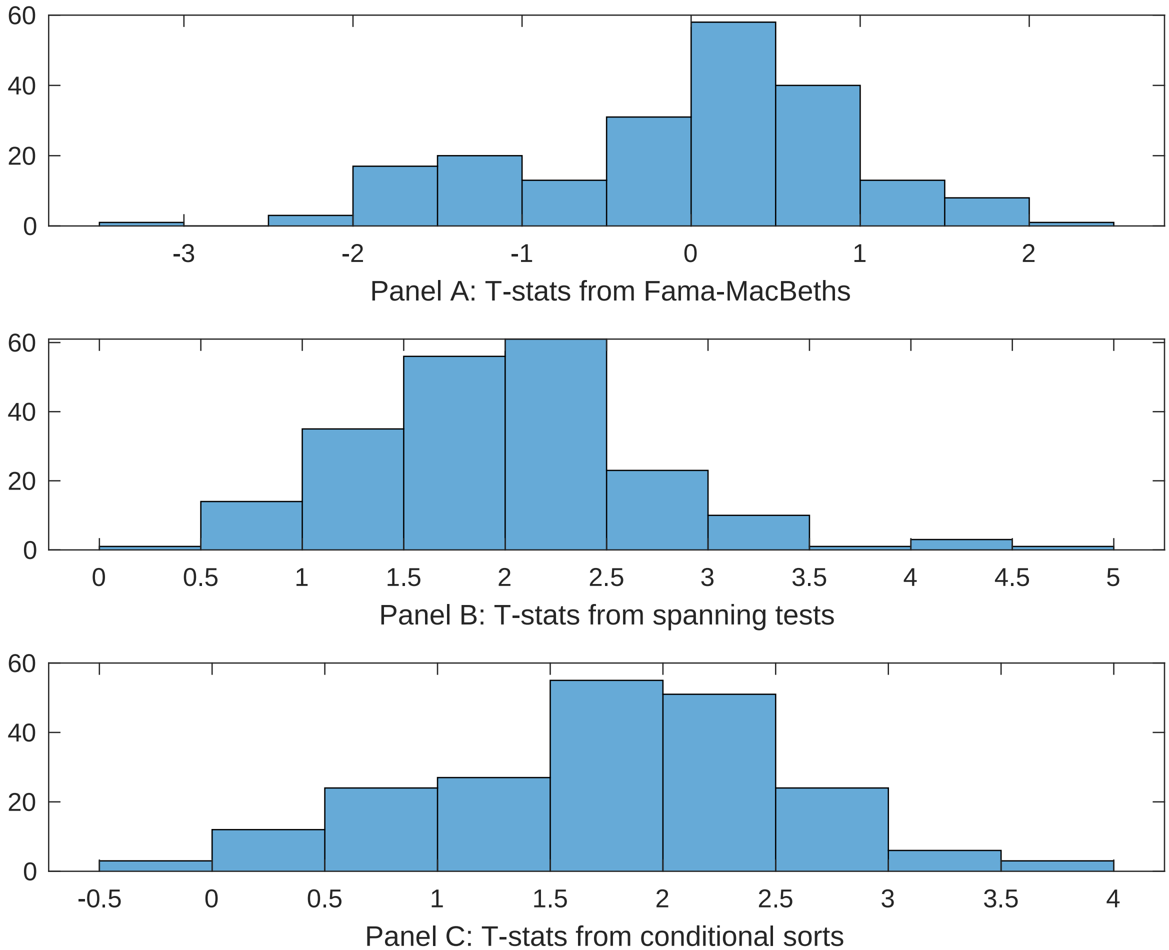


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES} PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on PES. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16	0.12	0.13	0.12	0.12	0.12	0.12
	[7.30]	[6.25]	[6.81]	[6.10]	[6.38]	[6.23]	[5.69]
PES	0.48	0.49	0.36	0.18	0.33	0.14	0.58
	[1.80]	[1.85]	[1.47]	[0.72]	[1.34]	[0.59]	[2.14]
Anomaly 1	0.43						0.58
	[3.59]						[0.40]
Anomaly 2		0.14					0.29
		[3.60]					[0.60]
Anomaly 3			0.84				0.59
			[7.11]				[4.76]
Anomaly 4				0.70			0.15
				[4.80]			[1.10]
Anomaly 5					0.15		0.50
					[1.62]		[0.56]
Anomaly 6						0.88	-0.30
						[1.86]	[-0.59]
# months	720	720	732	713	727	732	708
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PES trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PES} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29	0.28	0.28	0.21	0.26	0.23	0.20
	[3.25]	[3.06]	[3.03]	[2.26]	[2.90]	[2.53]	[2.18]
Anomaly 1	-27.56						-44.05
	[-5.64]						[-6.09]
Anomaly 2		-5.16					27.58
		[-1.13]					[4.13]
Anomaly 3			-19.20				-2.95
			[-3.53]				[-0.44]
Anomaly 4				-20.76			-13.62
				[-4.13]			[-2.67]
Anomaly 5					-13.76		-9.28
					[-5.21]		[-3.44]
Anomaly 6						-14.77	-11.48
						[-3.42]	[-2.42]
mkt	4.13	5.13	5.01	4.54	5.65	5.57	3.51
	[1.92]	[2.34]	[2.30]	[2.07]	[2.62]	[2.55]	[1.65]
smb	6.28	5.43	6.90	4.60	11.33	4.93	10.04
	[2.02]	[1.71]	[2.17]	[1.45]	[3.45]	[1.56]	[3.10]
hml	-4.64	-7.38	-7.38	-5.04	-3.36	-2.36	3.06
	[-1.11]	[-1.72]	[-1.75]	[-1.17]	[-0.79]	[-0.52]	[0.68]
rmw	42.34	43.17	43.93	42.99	37.77	44.82	40.32
	[10.09]	[10.04]	[10.34]	[10.08]	[8.75]	[10.52]	[9.44]
cma	-50.77	-72.28	-54.29	-58.02	-66.81	-69.20	-31.78
	[-6.58]	[-9.33]	[-6.02]	[-7.60]	[-10.47]	[-10.42]	[-3.35]
umd	5.36	4.73	4.27	5.70	4.07	3.90	5.44
	[2.51]	[2.17]	[1.97]	[2.61]	[1.91]	[1.79]	[2.56]
# months	732	732	732	713	728	732	709
$\bar{R}^2$(%)	46	44	45	45	46	45	50

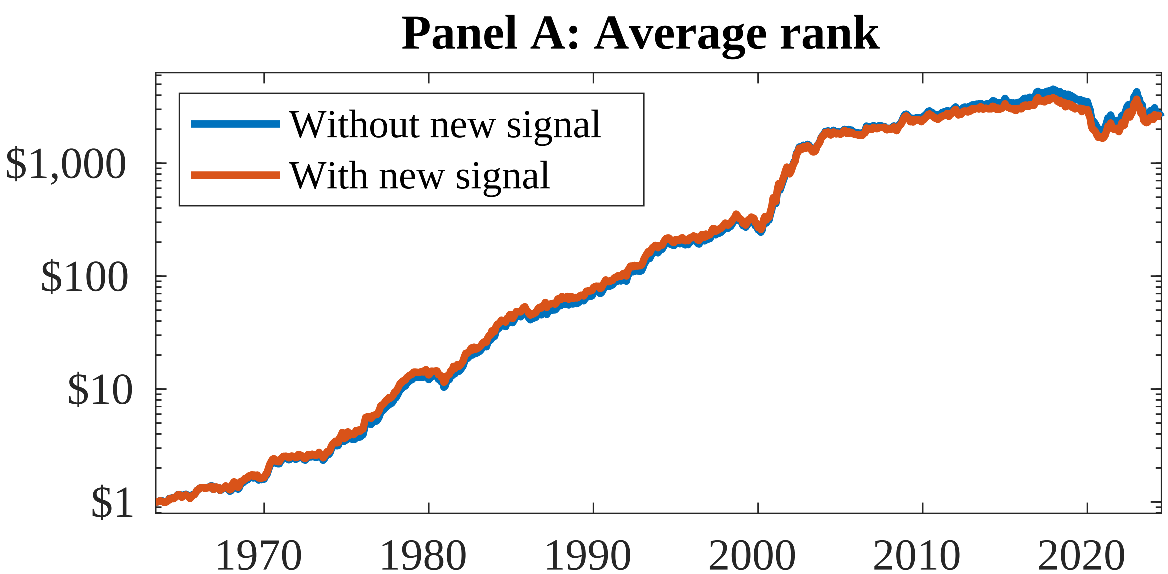


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PES. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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